EXNO: 9

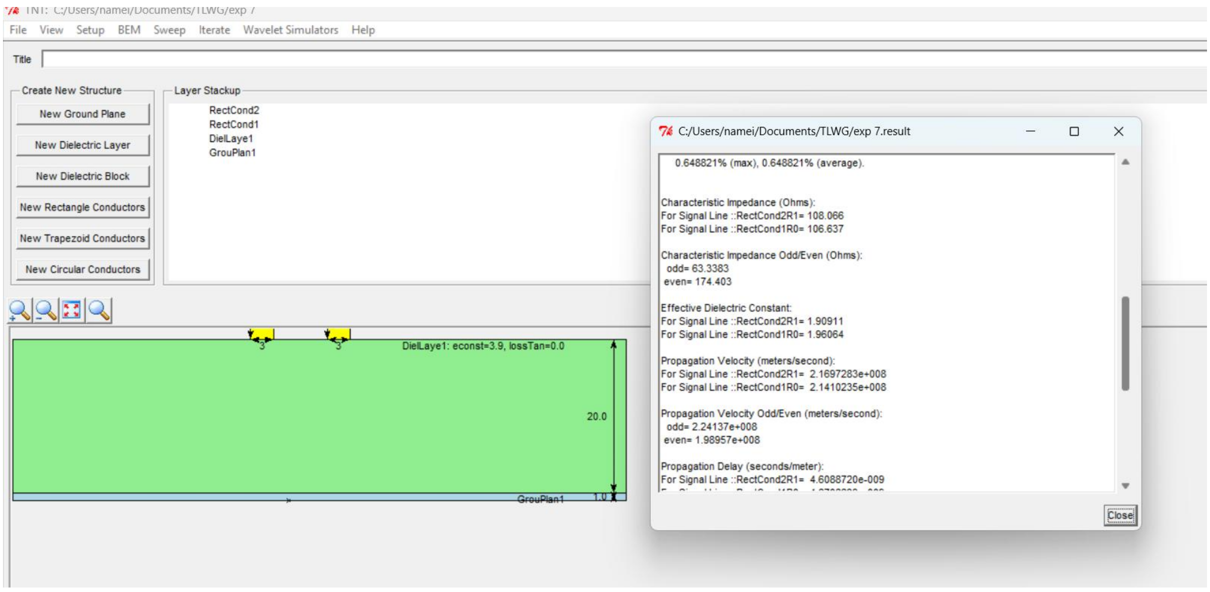
Design a two conductor transmission line and examine the impact of Dielectric layer material on characteristic Impedance and Propagation Velocity.

Test Case 1:

Design a Two conductor transmission line with single ground plane and investigate the impact of the Dielectric layer material on characteristic Impedance and Propagation Velocity. Materials to be tested - GETEK ,FR 404, ADS995 and RO3203 .

Output:

Dielectric Material	Permittivity	Loss Tangent	Simulation Z_0 (Ω)	Simulation Velocity (m/s)	Theoretical Z_0 (Ω)	Theoretical Velocity (m/s)
GETEK	3.9	0.008	107.00	1.816×10^8	107.00	1.816×10^8
FR 404	4.1	0.020	104.36	1.771×10^8	104.36	1.771×10^8
ADS995	9.8	0.0001	67.48	1.145×10^8	67.48	1.145×10^8
RO3203	3.02	0.0016	121.60	2.063×10^8	121.60	2.063×10^8



Test Case 2:

Design a Two conductor transmission line with single ground plane and investigate the impact of the conductor material on Rdc parameters. Width of the conductor is 18mm and the height is 1.4 mm.Conductor is placed on a 10 mm thickness dielectric layer.

Output:

Conductor Material	Conductivity (S/m)	Rdc R1 R1 (Ω/m)	Rdc R1 R0 (Ω/m)	Rdc R0 R1 (Ω/m)	Rdc R0 R0 (Ω/m)
Tungsten	1.79×10^7	**0.004434**	**0.002217**	**0.002217**	**0**
Aluminium	3.50×10^7	**0.002268**	**0.001134**	**0.001134**	**0**
Copper	5.80×10^7	**0.001368**	**0.000684**	**0.000684**	**0**
Silver	6.30×10^7	**0.001260**	**0.000630**	**0.000630**	**0**

Simulation Output: